

CSC 244 – Discrete Math for Computer Science II

Fall 2024

**The Commons, Rm 105
MWF 2:00-2:50 PM**

Course Description

This course builds on what was learned in 144 to further the student's knowledge about the mathematics used in computer science. The student learns how to form logical arguments and then to express them as clear proofs. Then the student learns advanced counting techniques depending on recurrence relations and proof by inductions. Combinatorics and probability are important parts of discrete mathematics. The students will learn how to apply them to solve different real-life problems. Graphs, which are used throughout computer science, are studied as well. Lastly, the course introduces computational theory. Finite automata and regular expressions are defined and explored.

Catalog Description:

A second course in mathematical concepts for computer science. Topics include: recurrence relations; proofs in propositional and first-order predicate logic; mathematical induction; graphs and trees; regular expressions and finite state machines.

Instructor and Contact Information

Instructor name: Reyan Ahmed, office location: GS 831, telephone number: (520) 621-4632, e-mail address: abureyanahmed@arizona.edu

Office Hours: Tuesdays and Thursdays 1:00 PM - 2:30 PM (any changes will be posted in D2L/Discord)

Teaching assistants: Leah Knodel (leahknodel@arizona.edu), Ankit Garg (ankitgarg@arizona.edu), Mahshad Habibpourparizi (mahshadh@arizona.edu), Tanishq Jaiswal (tjaiswal25@arizona.edu), Aaron Lam (aaronlam168@arizona.edu), Daniel Martinez (dmartinez5@arizona.edu), Cesar Quihuis-Romero (qromeroc@arizona.edu), Yashi Gupta (yashigupta@arizona.edu); TA office hours will be informed later

D2L course site: <https://d2l.arizona.edu/d2l/home/1487819>

Course Format and Teaching Methods

Besides the scheduled lectures there will be no lab. However, in the lecture sessions there will be in class activities, for example, often after describing a concept, the instructor will ask questions to the students; based on the answer the instructor will decide how to proceed to the next topic. Also, one activity of this course is the quizzes. The quiz questions will be provided during the lectures, and the students need to submit them in a short period of time. These are some reasons to attend the lectures. Absence in the lecture sessions without an unavoidable reason is not allowed. If for any unavoidable reason, you miss any lecture, go through the recordings, slides, and readings. If you still face any trouble contact the instructor. Besides class activities, there will be homeworks almost each week and exams as discussed below.

Course Objectives

- Learn the different structures of proofs.
- Learn to write correct, well formatted proofs.
- Learn how to use, solve, and prove recurrence relations.
- Count items in various contexts using the principles of combinatorics, including the sum and product rules, permutations and combinations, and the pigeonhole principle.
- Compute the probability of events and the expectations of random variables; determine if events are independent or dependent; identify binomial distributions and compute related probabilities; and compute variance.
- Learn basic terminology and structure of graphs and trees.
- Learn special cases of graphs and trees and practical algorithms used with them.
- Learn the basics of Regular Languages and how they can be represented by DFAs and Regular Expressions.
- Learn the basics of finite state machines and get an idea of computational complexity.

Expected Learning Outcomes

Students will be able to:

1. Solve a variety of basic recurrence relations.
2. Analyze a problem to determine underlying recurrence relations.
3. Apply formal logic proofs and/or informal, but rigorous, logical reasoning to real problems, such as predicting the behavior of software or solving problems such as puzzles.
4. Explain the parallels between ideas of mathematical and/or structural induction to recursion and recursively defined structures.
5. Explain the relationship between weak and strong induction and give examples of the appropriate use of each.
6. State the well-ordering principle and its relationship to mathematical induction.
7. Answer combinatorial questions algorithm complexity.
8. Apply counting to determine whether there are enough telephone numbers or Internet protocol addresses to meet demand.
9. Use pigeonhole principle, permutations and combinations to solve different counting problems.
10. Apply probability theory to determine the probability of an event.
11. Compute expected value and variance of simple scenarios.
12. Understand the application of probability theory in computer science such as algorithm design, proving claims, and spam detection.
13. Illustrate by example the basic terminology of graph theory, as well as some of the properties and special cases of each type of graph/tree.
14. Model a variety of real-world problems in computer science using appropriate forms of graphs and trees, such as representing a network topology or the organization of a hierarchical file system.
15. Show how concepts from graphs and trees appear in data structures, algorithms, proof techniques (structural induction), and counting.
16. Determine if two graphs are isomorphic.
17. Design a deterministic finite state machine to accept a specified language.
18. Generate a regular expression to represent a specified language.

Makeup Policy for Students Who Register Late

Students who register after the first class meeting due to some unavoidable reasons may make

up missed assignments/quizzes. The student may take an equal amount of time that other students have taken. *The students must communicate with the instructor and can only make up missed assignments if approved by the instructor.*

Course Communications

There will be various ways that communication will take place in this course, and it is important that you keep track of all of them.

- **The Syllabus:** You are expected to be familiar with the policies in the syllabus. If you have a question about the course expectations and policies, this is the first place you should look for an answer.
- **Lectures:** Sometimes things are announced in class, so if you are not present in class, make sure you watch the videos within 48 hours of the class time.
- **Email:** If I need to contact you individually, I will do so through your UA email, so make sure you check that regularly.
- **D2L:** Announcements may be made on D2L, so make sure you check that page regularly. I recommend checking both the announcements on a daily basis (at least during the week). You can also keep up with your grades on D2L.
- **Discord:** Announcements may be made on Discord, which is also used for questions and discussions *about course content*. **Please do not use Discord to ask about grades or request a regrade.** Make sure you add yourself and check regularly for announcements. This is especially important while assignments are out as any updates or clarifications about the assignment will be announced on Discord (<https://discord.com/invite/ZWt3aJDdu>).
- **Gradescope:** This site is where you will submit many of your assignments, and you should use this site for checking graded work and submitting regrade requests for those assignments. Exams will also be graded on Gradescope. A link to Gradescope is here: <https://www.gradescope.com>. You should be added to the course automatically, but if you are not, let me know, and I will sync the roster again.

Required Texts or Readings

We are using an inclusive access text for this course. In D2L, go to Content→ VitalSource Inclusive Access→eTextBook→Read Now. You should also have received an email regarding the book.

The book is: *Discrete Mathematics and Its Applications*, Eighth Edition, by Kenneth H. Rosen, ISBN # 978-1-259-67651-2.

Required or Special Materials

Some of the in-class activities may require a phone or another device that can access the Internet. However, these particular assignments are not for regular points.

Assignments and Examinations: Schedule/Due Dates

Quizzes/short Assignments (10 points)

There will be weekly quizzes/short assignments. These are not in the schedule, so you are encouraged to come to class regularly. *Submitting these assignments will be a lot easier if you attend the class in person, and they cannot be made up for any reason.* There will be 11-12 points possible in this category, and you can earn up to 10 points in the category. Note that some of these assignments may be graded for completion, and some may be graded for accuracy.

Homework (40 points)

There will be 10-12 written homework assignments, each worth 4 points. They must be submitted

on Gradescope as a single PDF, and they will not be accepted late for any reason. **Do not submit separate images on Gradescope. You can put images of your work into a single PDF and submit that, but it must be a single PDF. If you do not follow these instructions, we will not automatically deduct any points. However, it is at your own risk as students tend to have more submission problems with the image option than with the PDF option.** Note that there are actually more than 40 points available in this category, and you can earn up to 40, so there is some flexibility in case you miss an assignment. In most cases, some homework problems will be graded for completion and some for accuracy. You should always check the feedback after the homework is graded; if your answer is correct, then there will be no feedback and you will get the full credit. The tentative homework release date, time and deadlines are provided below.

Midterms (30 points)

There will be two midterms given during class. They are already scheduled (see the schedule below), so please plan accordingly. These are in-person exams and makeups will only be given in the case of an emergency that is verified by an official letter from the Dean of Students Office. Each midterm is worth 15 points.

Final (20 points).

The final exam will be on Friday, December 13, 2024 from 1:00-3:00 PM. It will be in person and comprehensive.

Extra Credit Opportunities. I sometimes provide various extra credit opportunities throughout the semester. Most of these are just announced as they come up, but one of them is explained below and is ongoing throughout the semester.

Each student has the opportunity to receive up to 2.5 extra credit points for “course engagement.” These points can be earned by

- *meaningful in-class participation:* Ask a good question, make an insightful comment, or answer a challenging question. (Note: Not every question/answer/comment will be awarded points, but if you are showing that you are actively engaged, you are likely to receive $\frac{1}{4}$ point.)
- *office hour participation:* To encourage you to keep up with the material and get help when you need it, you can earn $\frac{1}{4}$ extra credit point just by showing up and participating in instructor/TA office hours. Participation must be active in some way (i.e. showing up for a minute and then leaving without engaging is not going to earn you any points). Even if you do not have specific questions, you can show up and “check in” regarding your understanding of the assignments and the course materials. Both I and the TAs will keep track of office hour attendance, and you can earn $\frac{1}{4}$ point per session.
- NOTE: You are limited to $\frac{1}{4}$ point per class/office hour session.
- NOTE: We do our best to keep track of all of this stuff, but mistakes do happen especially when office hours get busy or there is a lot going in class. If you believe you have not received credit for something, please let us know. For example, if you participate in class, then let me know right after class so that I can write down your name.

Final Examination

The final exam will be on Friday, December 13, 2024 from 1:00-3:00 PM.

Regulations and Final Exam Schedule:

<https://registrar.arizona.edu/faculty-staff-resources/room-class-scheduling/schedule-classes/final-exams>

Grading Scale and Policies

Final grades will be determined by the number of points you earn throughout the course. The following lists the number of points possible in each category.

Category	Points Possible	Points Available
In-class Assignments	10	11-12
Homework (HW)	40	40-50
Midterms (MT)	30	30+
Final Exam (FE)	20	20+

Note: + indicates "at least"

The grading scale is given below. Adjustments to this scale may be made at the end of the semester, but you should not depend upon that. Any adjustments would lower grade cutoffs. No grade cutoffs will be raised.

Total Points	Letter Grade
90+	A
80-89.99	B
70-79.99	C
60-69.99	D
0-59.99	E

University policy regarding grades and grading systems is available at

<https://catalog.arizona.edu/policy/courses-credit/grading/grading-system>

Grading Times

- The quizzes will be due at the end of the day.
- The weekly homeworks will be due in a week, there are some exceptions as discussed below.
- Both quizzes and weekly homeworks will be graded in a week. For example, if the assignment/homework is due on Monday, 22nd January, 2024, then it will be graded by the following Monday, 29th January, 2024.
- The assignments and homeworks will be released in a fashion that the students will get them back before the corresponding midterm exams. The midterm exams will be graded in a week

as well. The final exam will be graded by the timeline provided by the university (usually 48 hours).

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at

<https://catalog.arizona.edu/policy/courses-credit/grading/grading-system>.

Dispute of Grade Policy

In order for us to keep the D2L gradebook accurate and up to date, it is important for any grade disputes or regrade requests to be done in a timely manner, so the deadline for making a regrade request on a particular assignment is usually 5 days after the grades are released.

Please note:

- *Discord* should never be used for regrade requests or any other discussion about grades.
- If an assignment is graded on Gradescope, you should submit a regrade request through Gradescope (one request per question).
- Please review any posted announcements before submitting a regrade request and make sure that your request clearly states why you believe the question was graded an error.

Honors Credit

Students wishing to contract this course for Honors Credit should e-mail me to set up an appointment to discuss the terms of the contract and to sign the Honors Course Contract Request Form. The form is available at <http://www.honors.arizona.edu/honors-contracts>

Scheduled Topic and Activities

This schedule is tentative. The official schedule is the schedule of Gradescope and will be updated as the semester progresses.

Date	Lecture Topics	Released	Due	Reading
M 8/26	Course Overview			
W 8/28	Introduction to Proofs			1.7
F 8/30	Proof Methods & Strategy			1.8
M 9/2	Labor Day - no classes			
T 9/3		HW 1		
W 9/4	Recurrence Relations			2.4
F 9/6	Recurrence Relations			2.4
M 9/9	Applications of Recurrence Relations	HW 2	HW 1	8.1
W 9/11	Solving Linear Recurrence			8.2

	Relations			
F 9/13	Divide-and-Conquer Algorithms and Recurrence Relations			8.3
M 9/16	Mathematical Induction	HW 3	HW 2	5.1
W 9/18	Strong Induction and Well-Ordering			5.2
F 9/20	Recursive Definitions and Structural Induction			5.3
M 9/23	Recursive Algorithms		HW 3	5.4
W 9/25	Program Correctness			5.5
F 9/27	Review/Buffer Day			
M 9/30	Midterm 1	HW 4		
W 10/2	The Basics of Counting			6.1
F 10/4	The Basics of Counting			6.1
M 10/7	The Pigeonhole Principle	HW 5	HW 4	6.2
W 10/9	Permutations and Combinations			6.3
F 10/11	Binomial Coefficients and Identities			6.4
M 10/14	Generalized Permutations and Combinations	HW 6	HW 5	6.5
W 10/16	Generating Permutations and Combinations			6.6
F 10/18	An Introduction to Discrete Probability			7.1
M 10/21	Probability Theory	HW 7	HW 6	7.2
W 10/23	Bayes' Theorem			7.3
F 10/25	Expected Value and Variance			7.4
M 10/28	Expected Value and Variance	HW 8	HW 7	7.4
W 10/30	Graphs & Graph Models			10.1

F 11/1	Graph Terminology & Special Types of Graphs			10.2
M 11/4	Representing Graphs & Graph Isomorphism		HW 8	10.3
W 11/6	Connectivity			10.4
F 11/8	Review/Buffer day			
M 11/11	Veterans Day - no classes			
W 11/13	Midterm 2	HW 9		
F 11/15	Euler & Hamilton Paths			10.5
M 11/18	Shortest-Path Problems			10.6
W 11/20	Planar Graphs	HW 10	HW 9	10.7
F 11/22	Introduction to Trees			11.1
M 11/25	Applications of Trees			11.2
W 11/27	Tree Traversal	HW 11	HW 10	11.3
F 11/29	Thanksgiving recess			
M 12/2	Spanning Trees			11.4
W 12/4	Minimum Spanning Trees	HW 12	HW 11	11.5
F 12/6	Regular Expressions			13.3
M 12/9	Finite-State Machines with No Output		HW 12	13.3
W 12/11	Review/Buffer Day			
F 12/13	Final (1 PM - 3 PM)			

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Students are asked to refrain from disruptive conversations with people sitting around them during lecture. Students observed engaging in disruptive activity will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave lecture or discussion and

may be reported to the Dean of Students.

Some learning styles are best served by using personal electronics, such as laptops and iPads. These devices can be distracting to other learners. Therefore, students who prefer to use electronic devices for note-taking during lectures should use one side of the classroom.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): <https://cirt.arizona.edu/case-emergency/overview>

Also watch the video available at

https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/app/me/ledetail;spf-url=common%2Flearningeventdetail%2Fcrty0000000000003841

University-wide Policies link

Links to the following UA policies are provided here: <https://catalog.arizona.edu/syllabus-policies>

- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy

Department-wide Syllabus Policies and Resources link

Links to the following departmental syllabus policies and resources are provided here, <https://www.cs.arizona.edu/cs-course-syllabus-policies> :

- Department Code of Conduct
- Class Recordings
- Illnesses and Emergencies
- Obtaining Help
- Preferred Names and Pronouns
- Confidentiality of Student Records
- Additional Resources
- Land Acknowledgement Statement

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.